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Lexical Analyzer Design: Melody Script-DSL

1. Introduction.

The Lexical Analyzer (Lexer) is the first phase of the Melody Script - DSL compiler. Its primary responsibility is to read the source code character by character and group them into meaningful sequences called tokens. This document outlines the design of Lexer, including the token categories and the transition table used for token recognition.

1. Introduction:

2. Token categories:

The MelodyScript Lexer identifies several categories of token.

Category	Description	Examples
Keywords	Reserved words with specific meanings in the languages	int, note, if, repeat.
Identifiers	User defined names for variables	bpm, myNoteCounter.
Literals	Constant value (numbers and musical notes)	120, C4, F#5
Operators	Symbol of arithmetic & logical operations	+, -, *, /, =
Punctuation	Structural Symbols.	(), { }, ; ,

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3. Transition Table for Token recognition
The following transition represents a simplified version of the Deterministic finite Automaton (DFA) logic used to recognize different token types.

Current Status	Input [a-zA-Z]	Input 0-9	Input A-G	Input # or b	Input symbol (+, -, etc)	Input =	Input white space.
State 1	State 1	State 1	State 1	Error	Accept ID/KW	Accept ID/KW	Accept ID/KW
State 2	Error	State 2	Error	Error	Accept NUM	Accept NUM	Accept NUM.
State 3	Error	State 6 (Note)	Error	State 7 (Note)	Accept Note if octave	Accept Note	Accept Note
State 4	Accept Symbol	Accept Symbol	Accept Symbol	Accept Symbol	Accept Symbol	Accept Symbol	Accept Symbol
State 5	Accept =	Accept =	Accept =	Accept =	Accept =	Accept =	State 8 (8) =
State 6	Error	Error	Error	Error	Accept note	Accept not	Accept not
State 7	Error	State 6 NOTE	Error	Error	Error	Error	Error
State 8	Accept =	Accept =	Accept =	Accept =	Error	Accept =	//

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4. DFA Diagram (Conceptual).

